BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: PR X (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** Mathematics Paper–X (01) (21)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. If $(x + 1)(x - 5) = 0$ then the solutions are.....

(A) $\{-1, 5\}$ (B) $\{-1, -5\}$

(C) $\{1, 5\}$ (D) $\{1, -5\}$

Q2. If a and b are the roots of $4x^2-3x+7=0$ then the value of $1/a + 1/b$

(A) $-3/4$ (B) $3/7$

(C) $-3/7$ (D) $4/7$

Q3. If $2a + 1 : 21 :: 4 : 7$, then a = ...

(A) 13/2 (B) 11/2

(C) 10 (D) 9/2

Q4. $(x^2 + 2)/(x^2 + 2x + 2)$ is a/an.....

(A) Proper fraction (B) Improper fraction

(C) Irrational fraction (D) None of these

Q5. If A has two elements and B has three elements, then number of binary relation in $A \times B$ is.....

(A) 2×3 (B) 2^3

(C) 2^6 (D) 2^2

Q6. If R = {(0,0) (8,2) (10,3) (14,12)} then Dom R = ...

(A) {0,8,10,14}
(B) {0,2,3,12}
(C) {8,10,4}
(D) {0,10}

Q7. The difference of the largest and smallest value in the data is called.....

(A) Mean
(B) Mode
(C) Range
(D) Standard Deviation

Q8. In the given set of data 63, 66, 65, 67, 69, median is.....

(A) 63
(B) 66
(C) 65
(D) 69

Q9. Sin(−350°) lies in.....

(A) 1st Quadrant
(B) 2nd Quadrant
(C) 3rd Quadrant
(D) 4th Quadrant

Q10. $1 + \tan^2\theta$ = ...

(A) $\sin^2\theta$
(B) $\cos^2\theta$
(C) $\text{cosec}^2\theta$
(D) $\sec^2\theta$

Q11. Apollonius was born in.....

(A) Perga
(B) Istanbul
(C) Islamabad
(D) Kabul

Q12. The sum of the roots of the equation $ax^2 + bx + c = 0$ is.....

(A) c/a
(B) −c/a
(C) b/a
(D) −b/a

Q13. In a circle, two chords are equally distant from the centre of the circle the chords are.....

(A) Congruent
(B) Not congruent
(C) Parallel
(D) Non parallel

Q14. The angle in a semi-circle is equal to.....

(A) 30°
(B) 45°
(C) 90°
(D) 180°

Q15. How many tangents can be drawn from a point outside the circle?

(A) 1
(B) 2
(C) 3
(D) 4

Section-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 36

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

QII. Attempt any NINE parts. Each part carries FOUR marks.

1. Solve $6x^4 - 13x^2 + 5 = 0$

2. Prove that $(1 + 2\omega)(1 + 2\omega^2)(1 - \omega - \omega^2) = 8$

3. Prove that synthetic division to find the value of K if "−2" is a zero of the polynomial $x^3 + 4x^2 + Kx + 8$

4. If α, β are the roots of $x^2 - 5x + k = 0$, find k such that $3\alpha + 2\beta = 12$

5. If $(x-y)/z = (y-z)/x = (z-x)/y$ then prove that: $x = y = z$ where x,y,z are non-zero numbers and $x + y + z \neq 0$

6. If $3x - 1$, 4, 35 are continued proportions, find the value of x.

7. Resolve $x^2/(x^3 + 2x + 1)$ into partial fractions.

8. If U = {1,2,3,.....,10}, A = {2,4,6,8,10}, B = {1,3,5,7,9}. Then prove that $(A \cup B)' = A' \cap B'$

9. Domain of a binary relation $R = \{(x,y) \mid y + 1 = 2x^2\}$ is set N. Find its range.

10. Find the Arithmetic Mean and Mode of the given data. 58 59 60 62 64 66 65 67 67 68 70 71 71 71 73

11. Prove that $\sin x/(1+\cos x) = (1-\cos x)/\sin x$

12. A tree is 50 m high. Find the angle of elevation of its top to a point, on the ground 100 m away from the foot of tree.

Section-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 24

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII. A straight line, drawn from the centre of a circle to bisect a chord (which is not diameter) is perpendicular to the chord.

QIV. The tangent to a circle and the radial segment joining the point of contact and the centre are perpendicular to each other.

QV. Perpendicular from the centre of a circle on a chord bisects it.

QVI. Draw a ΔABC with $m\angle A = 60°$ and $m\angle B = 45°$. Draw three angle bisectors and then inscribe a circle in it.